

About Me

Hello! I'm **Kaviyadharshini M**, a third-year Computer Science and Engineering student with a strong passion for **Artificial Intelligence, Machine Learning, Data Science, and Software Development**. I am enthusiastic about leveraging technology to solve real-world problems and creating intelligent applications that make a meaningful impact.

My interest in technology began with programming and problem-solving. As I explored Python and data analysis, I became increasingly fascinated by how machines can learn from data and assist humans in making better decisions. This curiosity led me to dive deeper into Machine Learning, Natural Language Processing (NLP), Generative AI, and modern AI systems such as Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG).

I enjoy understanding not only how a technology works but also why it works. Rather than relying solely on tutorials or pre-built frameworks, I prefer implementing concepts from scratch, experimenting with different approaches, analyzing results, and continuously refining my solutions. This practical learning mindset has helped me strengthen both my technical and analytical skills.

Educational Background

I am currently pursuing a **Bachelor of Engineering in Computer Science and Engineering** at **Rathinam Technical Campus**.

My academic journey has provided me with a solid understanding of fundamental computer science concepts, including:

- **Programming and Software Development**
- **Data Structures and Algorithms**
- **Database Management Systems**
- **Artificial Intelligence**
- **Machine Learning**
- **Data Science**

Through coursework, laboratory sessions, internships, and personal projects, I have developed a balance between theoretical knowledge and practical implementation.

I believe that a strong foundation in programming, algorithms, databases, and data-driven technologies is essential for building scalable software and intelligent AI-powered systems.

Technical Skills

Programming Languages

I have experience working with:

- Python
- Java
- SQL

Python is my primary programming language for Data Science, Machine Learning, and AI-related projects. I use Java for object-oriented programming concepts and SQL for querying, managing, and analyzing structured data.

Data Science & Analytics

I possess hands-on experience with:

- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

My experience includes:

- Data Collection
- Data Cleaning
- Data Preprocessing
- Exploratory Data Analysis (EDA)
- Data Visualization
- Feature Engineering
- Feature Selection
- Model Training
- Model Evaluation
- Hyperparameter Tuning
- Predictive Analytics

I enjoy transforming raw datasets into meaningful insights through analysis and visualization.

Artificial Intelligence & Machine Learning

My key areas of interest include:

- Machine Learning
- Deep Learning
- Natural Language Processing (NLP)
- Large Language Models (LLMs)
- Retrieval-Augmented Generation (RAG)
- Vector Databases
- Prompt Engineering
- AI Agents
- Multi-Agent Systems
- AI Workflow Automation
- Generative AI

I am particularly interested in understanding how modern AI applications integrate retrieval systems, embeddings, vector search, and language models to generate accurate and context-aware responses.

Tools & Technologies

I have worked with:

- Git & GitHub
- Streamlit
- FAISS
- LangChain
- Google Gemini API
- Jupyter Notebook
- VS Code
- PostgreSQL

- React.js
- Node.js
- YOLOv8

These tools have helped me build end-to-end applications, from data processing and model development to frontend deployment.

Internship Experience

Data Science Intern – Adventure Technology Solutions

During my internship, I gained practical exposure to industry-level data science workflows.

Key Responsibilities

- Cleaning and preprocessing large datasets
- Handling missing and inconsistent data
- Performing Exploratory Data Analysis
- Building machine learning models
- Conducting feature engineering
- Improving data quality
- Assisting in predictive analytics tasks
- Supporting business decision-making through data insights

Key Learnings

This internship strengthened my understanding of:

- Real-world data challenges
 - Data preprocessing pipelines
 - Model development lifecycle
 - Data-driven problem solving
 - Collaboration within technical teams
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Data Science Intern – Internboot

At Internboot, I focused on data analytics and visualization.

Responsibilities

- Data preprocessing
- Data cleaning
- Exploratory Data Analysis
- Dashboard development
- Data visualization
- Business reporting

Skills Developed

- Presenting technical insights visually
- Communicating findings effectively
- Creating meaningful reports
- Understanding business-oriented analytics

This experience enhanced both my technical and communication skills.

Academic & Personal Projects

Resume RAG Assistant

The Resume RAG Assistant is a Retrieval-Augmented Generation system that enables users to ask questions about resume documents and receive context-aware responses.

Technologies Used

- Python
- Sentence Transformers
- FAISS
- Gemini API

Features

- PDF text extraction
- Recursive text chunking
- Semantic search
- Vector embeddings
- Similarity retrieval
- Context-aware answer generation

What I Learned

Through this project, I gained hands-on experience with:

- Embedding models
 - Vector databases
 - Semantic search
 - Information retrieval
 - RAG architecture
 - LLM integration
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Agri Chatbot

The Agri Chatbot is an intelligent question-answering system designed to help users obtain agriculture-related information quickly and accurately.

Technologies

- Python
- Streamlit
- LangChain
- FAISS

Features

- User-friendly interface
- Semantic document retrieval
- Context-aware responses
- Knowledge-based question answering

Learning Outcomes

- Building conversational AI systems
 - Retrieval pipelines
 - LangChain integration
 - Prompt engineering
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Pothole Detection System

The Pothole Detection System uses computer vision techniques to identify potholes from road images.

Technologies

- YOLOv8
- React.js
- Node.js

Objectives

- Detect potholes automatically
- Improve road maintenance efficiency
- Reduce manual inspection efforts

Skills Gained

- Object detection
- Computer vision
- Dataset preparation
- Frontend-backend integration

Learning Approach

I strongly believe that practical implementation is the most effective way to learn technology.

Whenever I encounter a new topic, I typically follow a structured learning process:

1. Understand the fundamental concepts.
2. Study real-world use cases.
3. Implement the concept from scratch.
4. Experiment with multiple approaches.
5. Debug errors and understand failures.
6. Optimize the solution.
7. Build a project around the concept.

This approach has enabled me to gain a deeper understanding of AI systems rather than simply using them as black boxes.

Strengths

Some of my key strengths include:

- Strong willingness to learn
- Consistency and discipline
- Analytical thinking
- Problem-solving ability
- Curiosity about emerging technologies
- Adaptability to new tools and frameworks
- Self-learning mindset

I enjoy challenging myself with new concepts and continuously improving my technical skills.

Career Aspirations

My short-term goal is to secure a role as an:

- AI Engineer
- Machine Learning Engineer
- Data Scientist
- NLP Engineer

I aspire to work on intelligent systems that combine machine learning, software engineering, and data-driven decision-making.

In the long term, I aim to contribute to innovative AI products and become an expert in:

- Generative AI
- Machine Learning
- Deep Learning
- AI Agents
- Retrieval-Augmented Generation
- Natural Language Processing
- AI Infrastructure

I also hope to contribute to open-source AI projects and stay actively involved in the rapidly evolving AI ecosystem.

Personal Interests

Beyond academics and technology, I enjoy:

- Building side projects
- Learning emerging AI technologies
- Reading technical articles and blogs
- Exploring open-source software
- Participating in coding challenges
- Practicing problem-solving
- Learning about new industry trends

These activities help me remain curious, motivated, and continuously engaged in learning.

Conclusion

I am passionate about combining Artificial Intelligence, Data Science, and Software Engineering to develop innovative solutions for real-world challenges. Every project, internship, and learning experience contributes to my growth as an engineer and helps me move closer to my goal of becoming a skilled AI professional.

I believe that continuous learning, practical implementation, and persistence are the keys to success in technology. As I continue my journey, I look forward to building impactful applications, solving meaningful problems, and contributing to the advancement of intelligent systems.